

Progressive gate-unfreezing of the QFT operator

One gate thawed per stage, trained to a plateau — three unfreeze orderings $\times$ two initialisations on DIV2K-8q

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Question

The **QFTBasis(8, 8)** (72 $U(2)$ gates) starts either at the QFT-family **identity** ($T_{t=0}(x) = x$) or **Haar-random**. We **unfreeze one gate per stage**, train it to a plateau, then thaw the next. Does the **order** — block-growth (bg), left $\rightarrow$ right (lr), right $\rightarrow$ left (rl) — or the **initialisation** change the trajectory or the endpoint?

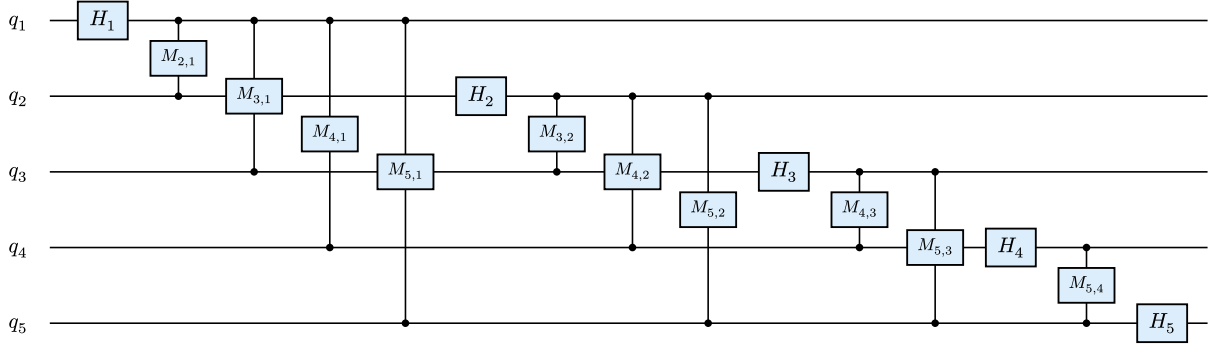


Figure 1: The QFT(5) circuit (one axis) with every gate named: H_i = Hadamard on q_i ; $M_{j,i}$ = controlled-phase between q_j and q_i ($j > i$). The three orderings thaw these same gates in the sequences below; the $m = 8$ runs use the analogous 72-gate circuit.

The three unfreeze orderings traverse the gates of Figure 1 as:

- **block-growth** (bg): finish QFT(k) before adding qubit $k\{+\}1$ — H_1 ; $H_2, M_{2,1}$; $H_3, M_{3,1}, M_{3,2}$; $H_4, M_{4,1}, M_{4,2}, M_{4,3}$; $H_5, M_{5,1}, \dots, M_{5,4}$. Long-range couplings come last.
- **left $\rightarrow$ right** (lr): QFT construction order — $H_1, M_{2,1}, M_{3,1}, M_{4,1}, M_{5,1}, H_2, M_{3,2}, \dots, H_5$.
- **right $\rightarrow$ left** (rl): the reverse — H_5 first, H_1 last.

Setup

Fixed batch of 50 training images; **MSELoss** on the top-10% coefficients. pdft's JIT Adam step runs with a per-stage frozen set (moments reset per stage); a Riemannian grad-norm probe gives the plateau signal. A stage ends at $\|g\| < 10^{-5}$ or $|\Delta L| < 10^{-5}$ (min 5 steps), else an 800-step cap. Random init: Haar $U(2)$ per Hadamard, uniform phase per controlled-phase, seed shared across orderings.

Training dynamics

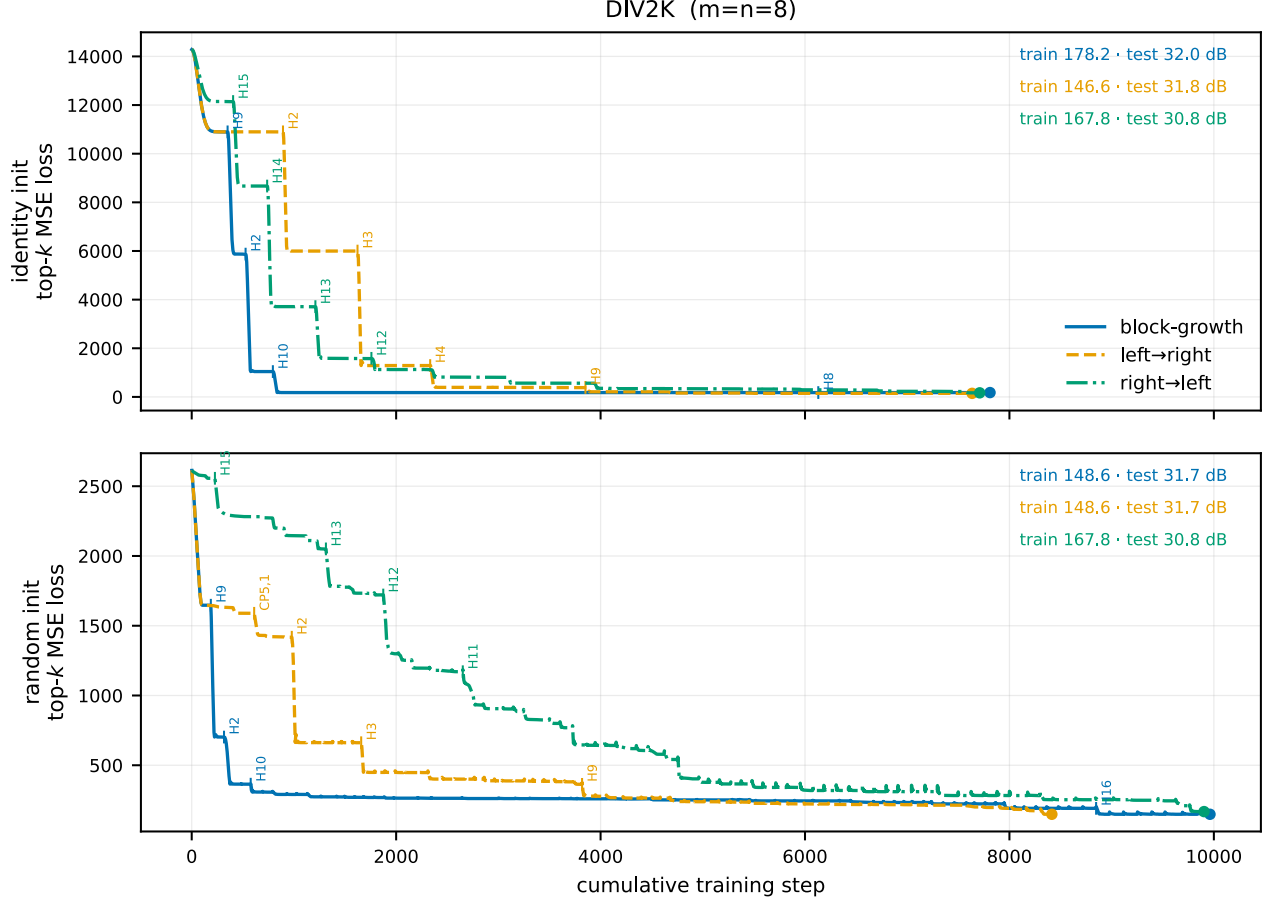


Figure 2: Absolute **training** top- k MSE loss vs cumulative step, one curve per ordering (rows: identity / random init). Tick labels mark **which gate** (e.g. H7, CP3,1) was thawed at each of the largest drops. Endpoints report **train MSE** and **test PSNR@ $\rho\{=\}.20$** — note these are **different** metrics (training top-10%-coefficient loss vs test-set image reconstruction keeping 20%), so a curve can sit higher in train loss yet score a higher test PSNR. Per-cell loss+grad-norm staircases: [div2k_8q/<init>/figures/](#).

Reconstruction PSNR (DIV2K)

Test PSNR (dB) at the final stage, with cumulative steps and plateau-trigger counts (grad-norm / loss- Δ / cap):

init	order	$\rho\{=\}.05$	$\rho\{=\}.10$	$\rho\{=\}.15$	$\rho\{=\}.20$	steps (g/ Δ /cap)
identity	block-growth	19.4	26.9	29.7	32	7810 (0/72/0)
identity	left→right	25.2	27.9	30	31.8	7635 (0/72/0)
identity	right→left	24.8	27.2	29.1	30.8	7709 (0/72/0)
random	block-growth	25.2	27.8	29.8	31.7	9962 (0/72/0)
random	left→right	25.2	27.8	29.8	31.7	8415 (0/72/0)
random	right→left	24.8	27.2	29.1	30.8	9904 (0/72/0)

Block-size baseline (comparison)

qft_progressive reaches the **same** QFT(8,8) on DIV2K via a **block-size** curriculum (train all $k(k+1)$ gates of a $2^k \times 2^k$ QFT at once, $k = 1..8$):

k	block	# gates	$\rho\{=\}.05$	$\rho\{=\}.10$	$\rho\{=\}.15$	$\rho\{=\}.20$
1	2×2	2	9.4	12.5	16.2	21.1
2	4×4	6	19.1	26.9	29.8	32

k	block	# gates	$\rho\{=\}.05$	$\rho\{=\}.10$	$\rho\{=\}.15$	$\rho\{=\}.20$
3	8×8	12	25.2	28.1	30.3	32.3
4	16×16	20	25.3	28	30.1	32
5	32×32	30	25.2	27.8	29.8	31.7
6	64×64	42	25.2	27.8	29.8	31.7
7	128×128	56	25.2	27.8	29.8	31.7
8	256×256	72	25.2	27.8	29.8	31.7

It saturates by $k=2$; the full QFT(8,8) endpoint is **31.7 dB** @ $\rho\{=\}.20$. The gate-unfreeze sweep lands at the same value (identity `bg/lr` 32/31.8 dB, random ≈ 31.7), so the QFT(8,8) optimum is **schedule-independent**. For reference, the best **any** prior trained basis reaches on DIV2K is 33.7 dB (`#pastmax.div2k_8q.by_rho.at("0.20").method`); the QFT family sits ≈ 2 dB under the richer U(4) families.

Reading

The staircase shows each gate’s marginal value: a tall drop is a gate that mattered (mostly the early, small-block gates — see the marked steps), a flat plateau a gate left near where it was thawed. All three orderings and both inits converge to ≈ 31.7 dB, so **order and curriculum set only the path, not the destination** — `rl` merely trails by ≈ 1 dB en route. The train-loss order need not match the test-PSNR order (`bg` has the **highest** train MSE yet the **best** test PSNR): the loss is a top-10%-coefficient proxy on the train batch, PSNR a 20%-keep reconstruction on the test set, so the two rank slightly differently in this near-flat optimum.